SOLUTION KEY: DESCRIPTION OF MOTION

Chapter 14 | Mechanics Unit

Part I: Conceptual Understanding

1. Multiple Choice Questions (MCQs)

a) **(ii) The train's seat** (The distance between the passenger and the seat does not change).

b) **(iii) Trajectory** (By definition, the path is the trajectory).

2. True or False

☐ **False** (Motion and rest are relative concepts).

☐ **True** (Displacement is an algebraic value; negative signifies direction opposite to the axis).

☐ **False** (Instantaneous speed is measured over a very narrow/short time interval).

3. Completion

- A **Frame of Reference** is an object used for comparison to determine if others are moving.
- In 1D motion, the position of a mobile M is determined by its **abscissa** x.
- The SI unit for acceleration is **m/s**^{2}.

Part II: Vectors and Representation

4. The Characteristics of Velocity

1. **Origin:** The current position M of the mobile.
2. **Line of Action:** The trajectory (the rectilinear axis).
3. **Direction:** The direction of motion.
4. **Magnitude (Intensity):** The instantaneous speed v at that moment.

Part III: Numerical Calculations

5. Speed and Duration

i. $\Delta x = x_f - x_i = 50 - (-10) =$ **60** m.

ii. $v_{av} = \frac{d}{\Delta t} = \frac{60}{4} =$ **15** m/s.

iii. v_{av}(km/h) $= 15 \times 3.6 =$ **54** km/h.

6. Nature of Motion

i. $a_{av} = \frac{v_2 - v_1}{\Delta t} = \frac{5-20}{3} = \frac{-15}{3} = -$**5** m/s**^{2}**.

ii. **Nature:** Decelerated Rectilinear Motion.
Justification: The velocity v is positive (assuming motion in the positive direction) while the acceleration a is negative. Since $v \cdot a < 0$, the motion is decelerated.

Part IV: Chronophotography (Advanced)

7. Formula Application & Acceleration Deduction *Given:* $\tau = 0.04$ *s*, $M_1M_3 = 8$ *cm* $= 0.08$ *m*, $M_3M_5 = 12$ *cm* $= 0.12$ *m.*

1. $v_2 = \frac{M_1M_3}{2\tau} = \frac{0.08}{2\times0.04} = \frac{0.08}{0.08} = \mathbf{1.0}$ m/s.
2. $v_4 = \frac{M_3M_5}{2\tau} = \frac{0.12}{2\times0.04} = \frac{0.12}{0.08} = \mathbf{1.5}$ m/s.
3. Formula for a_3: $a_3 = \frac{v_4-v_2}{t_4-t_2} = \frac{\mathbf{v_4-v_2}}{\mathbf{2\tau}}$.
4. Calculation: $a_3 = \frac{1.5-1.0}{2\times0.04} = \frac{0.5}{0.08} = \mathbf{6.25}$ m/s$^{\mathbf{2}}$.
5. **Nature:** Accelerated Rectilinear Motion.
   Justification: The speed is increasing ($v_4 > v_2$), and v and a have the same sign ($v \cdot a > 0$).